

Correlation and Regression— Moving Together

[Code ▾](#)

Foundational Concepts

In everyday life, social situations, you have lots of examples of cause and effect. If a student studies more, that student may possibly do better on the next test. The income of a person is related to the amount of education that they have. But you could certainly ask the question of whether the increase in the income is a result of the knowledge or skills they gained from education.

[Hide](#)

```
if (!caracas::has_sympy()) {  
  caracas::install_sympy()  
}  
library(caracas)  
library(sets)  
library(tibble)  
library(ggplot2)  
library(dplyr)  
library(broom)  
library(RColorBrewer)
```

[Hide](#)

```
def_sym("X", "Y")  
  
cmStats <- modules::use("CmStats.R")$cmStats
```

[Hide](#)

```
cmStats$μ(1:10)
```

```
[1] 5.5
```

[Hide](#)

```
Xs = c(0, 5, 20, 3, 15, 8)  
Ys = c(2, 10, 10, 6, 25, 4)  
  
sort(Xs)[(6 + 1) / 2]
```

```
[1] 5
```

[Hide](#)

```
median(sort(Xs))
```

```
[1] 6.5
```

[Hide](#)

```
caracas::as_func(cmStats$P_sym)
```

```
function(Evn, p)
{
  Evn/p
}
<environment: 0x31ebfcbe0>
```

[Hide](#)

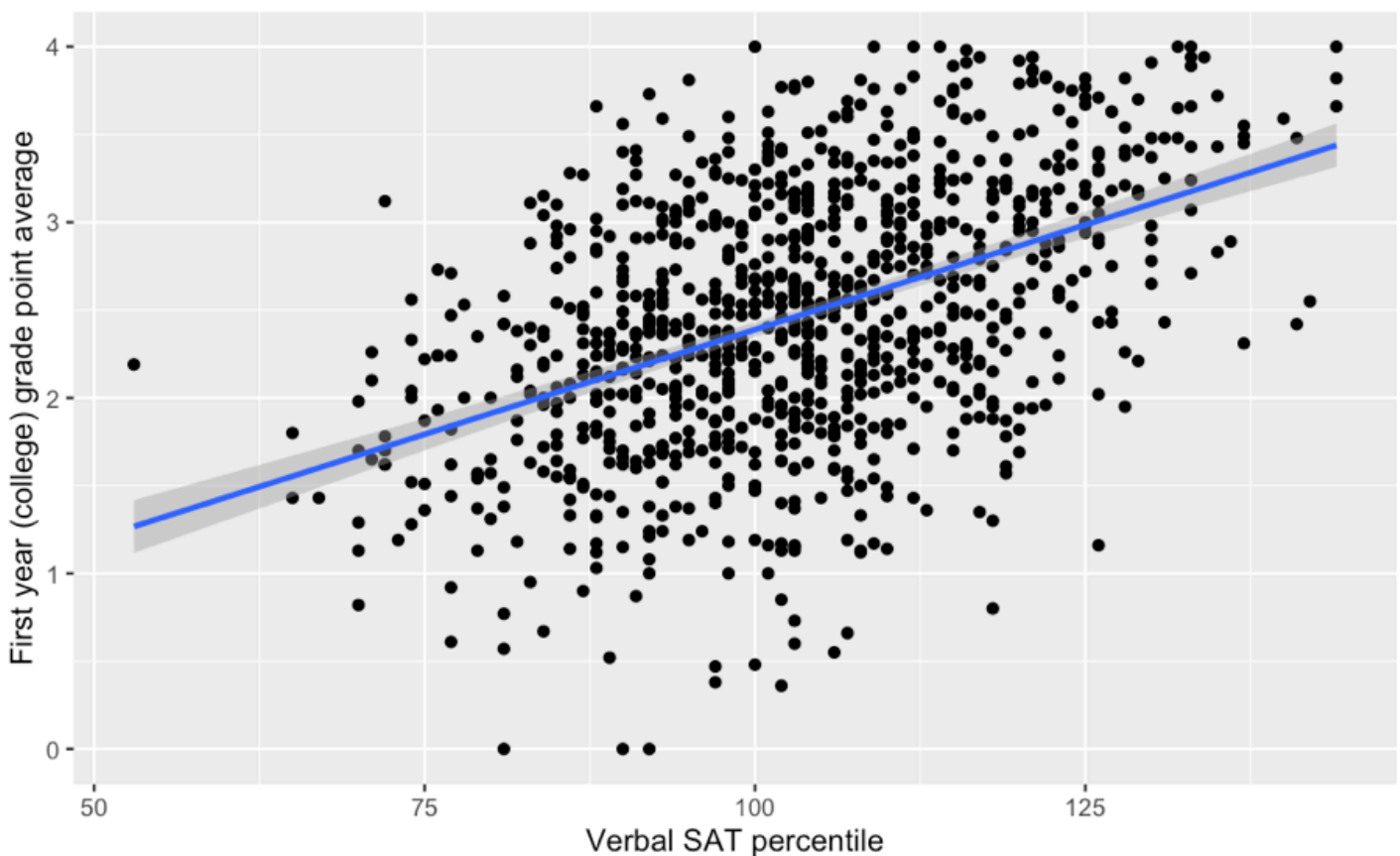
```
cmStats$P(1, 6)
```

```
[1] 0.1666667
```

Statistical graphs are useful in helping us visualize data. Through graphs, we understand data properties, such as the mean, median, and standard deviation; find patterns in data, such as clustering and correlation; suggest an underlying model that could have generated the data; verify our assumptions; check and fortify our analysis; summarize; and communicate results. This lecture will define and identify basic summaries of data, both numerical and graphical, and use R for calculating descriptive statistics, making graphs, and even writing functions that work on multiple datasets.

[Hide](#)

```
ggplot(satgpa, aes(x = sat_sum, fy_gpa)) +
  geom_point() +
  geom_smooth(method = "lm") +
  labs(
    x = "Verbal SAT percentile",
    y = "First year (college) grade point average"
  )
```



Describing relationships and cause and effect among variables is a basic strategy for understanding the world. A statistical challenge is to describe and measure how two variables, such as incoming SAT scores and college GPAs, are related. Higher SAT scores correspond with better GPAs in college; however, the SAT scores do not cause higher grades. In this lecture, we will introduce the ideas of scatter plots, which give a visual sense of the relationships between two variables; correlation, which gives a quantitative measure of the strength of the linear relationship between two varying quantities; and linear regression, which produces a straight-line approximation for a set of paired variables.

Hide

```
cor(satgpa$sat_sum, satgpa$fy_gpa)
```

```
[1] 0.460281
```

We can quantify the extent to which two variables are related by giving a number defining the correlation that summarizes the relationship. If one quantity increases and the other quantity tends to increase, the correlation is called positive, and the opposite is negative. Unlike the scatter plot of SAT scores and GPAs, which demonstrates a positive correlation, the scatter plot of two negatively correlated quantities goes down as it goes to the right.

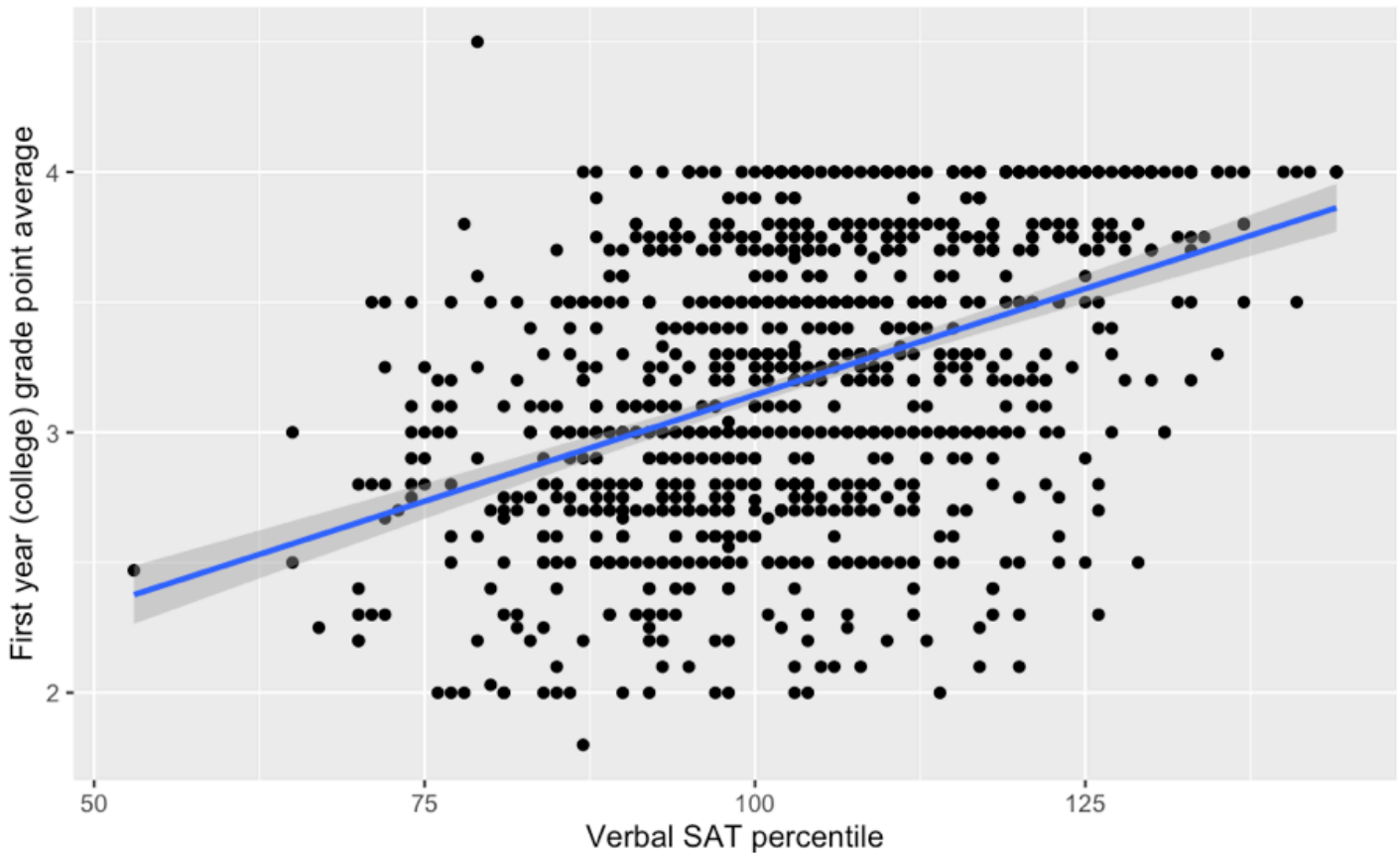
Hide

```
cor(satgpa$sat_v, satgpa$sat_m)
```

```
[1] 0.4665806
```

Hide

```
ggplot(satgpa, aes(x = sat_sum, hs_gpa)) +
  geom_point() +
  geom_smooth(method = "lm") +
  labs(
    x = "Verbal SAT percentile",
    y = "First year (college) grade point average"
  )
```



Hide

```
cor(satgpa$hs_gpa, satgpa$fy_gpa)
```

```
[1] 0.5433535
```

The correlation is denoted r (or [the Greek letter rho]). It is computed as follows: For each member of the population (e.g., student), take the distance of the rst variable (SAT score) from the rst variable's mean in units of standard deviation (that is, take the z-score of the SAT score) multiplied by the distance of the second variable (GPA) from its mean, again measured in units of standard deviation (that is, multiply by the z-score of the GPA).

Hide

```
cmStats$rho_sym(X, Y)
```

$$c: \frac{(X - \mu) \cdot (Y - \mu)}{\sigma \cdot (n - 1)}$$

We often summarize a scatter plot by a straight line. If the data set has a high correlation, the straight line will lie near most of the points. The straight line is a kind of summary of the data. The line is called a linear regression line. It is a line that approximates the data.

Hide

```
c(sd(satgpa$sat_sum), mean(satgpa$sat_sum))
```

```
[1] 14.28737 103.32900
```

Hide

```
c(sd(satgpa$fy_gpa), mean(satgpa$fy_gpa))
```

```
[1] 0.7408052 2.4679500
```

Similar mechanisms are used when there are more than two variables. Several explanatory variables can often make better predictions than any single variable alone. When two or more variables are used to predict another variable, we use multiple regression to summarize the situation.

Hide

```
cor(satgpa)
```

	sex	sat_v	sat_m	sat_sum	hs_gpa	fy_gpa
sex	1.00000000	-0.03915976	-0.2523657	-0.1718271	0.1410078	0.1002417
sat_v	-0.03915976	1.00000000	0.4665806	0.8522618	0.3595001	0.4013904
sat_m	-0.25236569	0.46658064	1.00000000	0.8603333	0.3780704	0.3871178
sat_sum	-0.17182706	0.85226183	0.8603333	1.00000000	0.4307883	0.4602810
hs_gpa	0.14100776	0.35950008	0.3780704	0.4307883	1.00000000	0.5433535
fy_gpa	0.10024168	0.40139036	0.3871178	0.4602810	0.5433535	1.00000000

Statistical correlation indicates an association but does not prove that there is a causal relationship between the variables. **One of the misuses of statistical information is to mistakenly infer cause and effect from correlation.**

Concept and Applications

VALIDATING STATISTICAL ASSUMPTIONS

Statistical graphs are useful in helping us visualize data. Through graphs, we understand data properties, such as the mean, median, and standard deviation; find patterns in data, such as clustering and correlation; suggest an underlying model that could have generated the data; verify our assumptions;

check and fortify our analysis; summarize; and communicate results. This lecture will define and identify basic summaries of data, both numerical and graphical, and use R for calculating descriptive statistics, making graphs, and even writing functions that work on multiple datasets.

BAR PLOTS

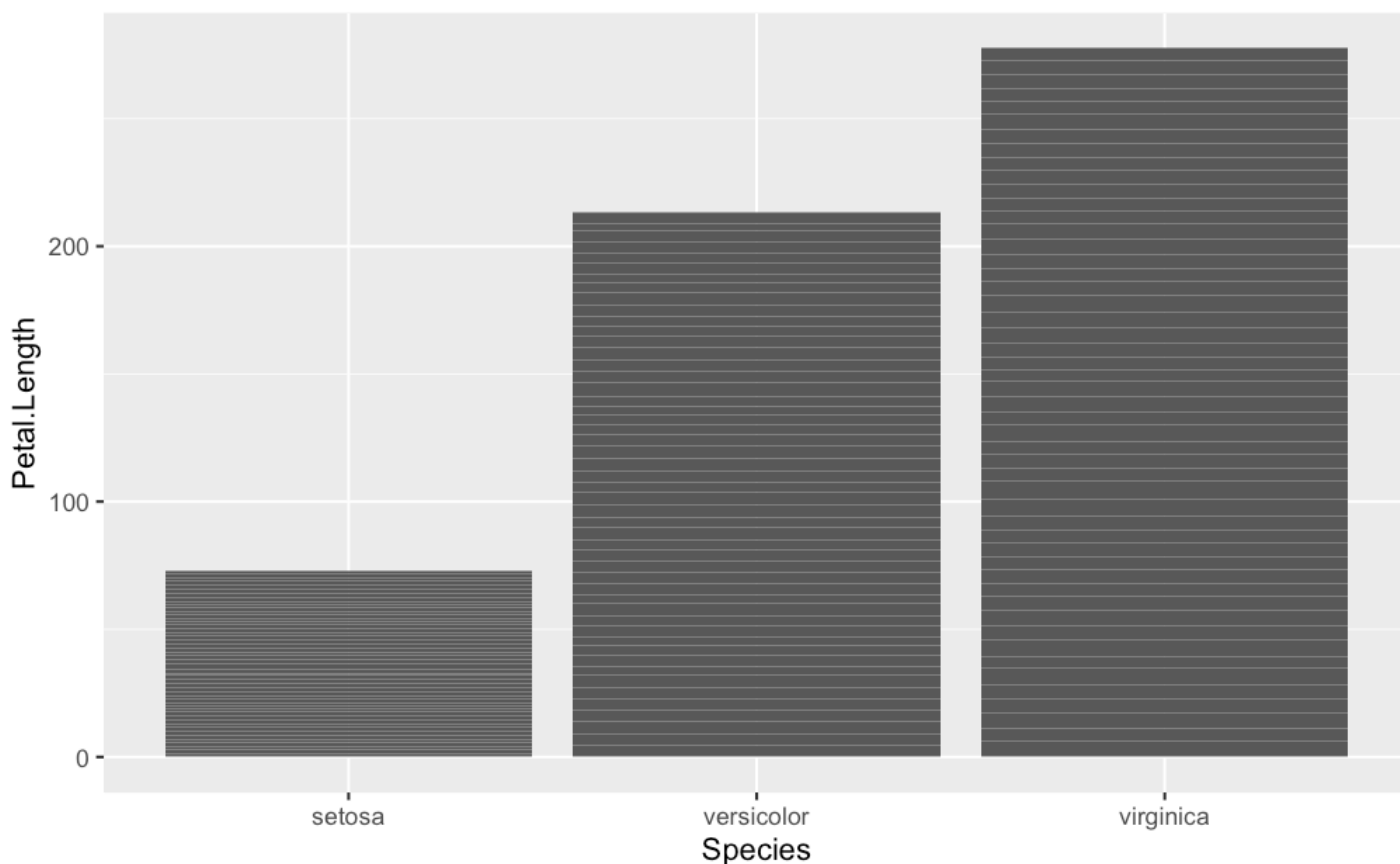
Hide

```
attach(iris)
```

The “iris” dataset is widely used throughout statistical science for illustrating various problems in statistical graphics, multivariate statistics, and machine learning. It’s a small but nontrivial dataset. The data values are real (as opposed to simulated) and are of high quality (collected with minimal error). The data were used by the celebrated British statistician Ronald Fisher in 1936.

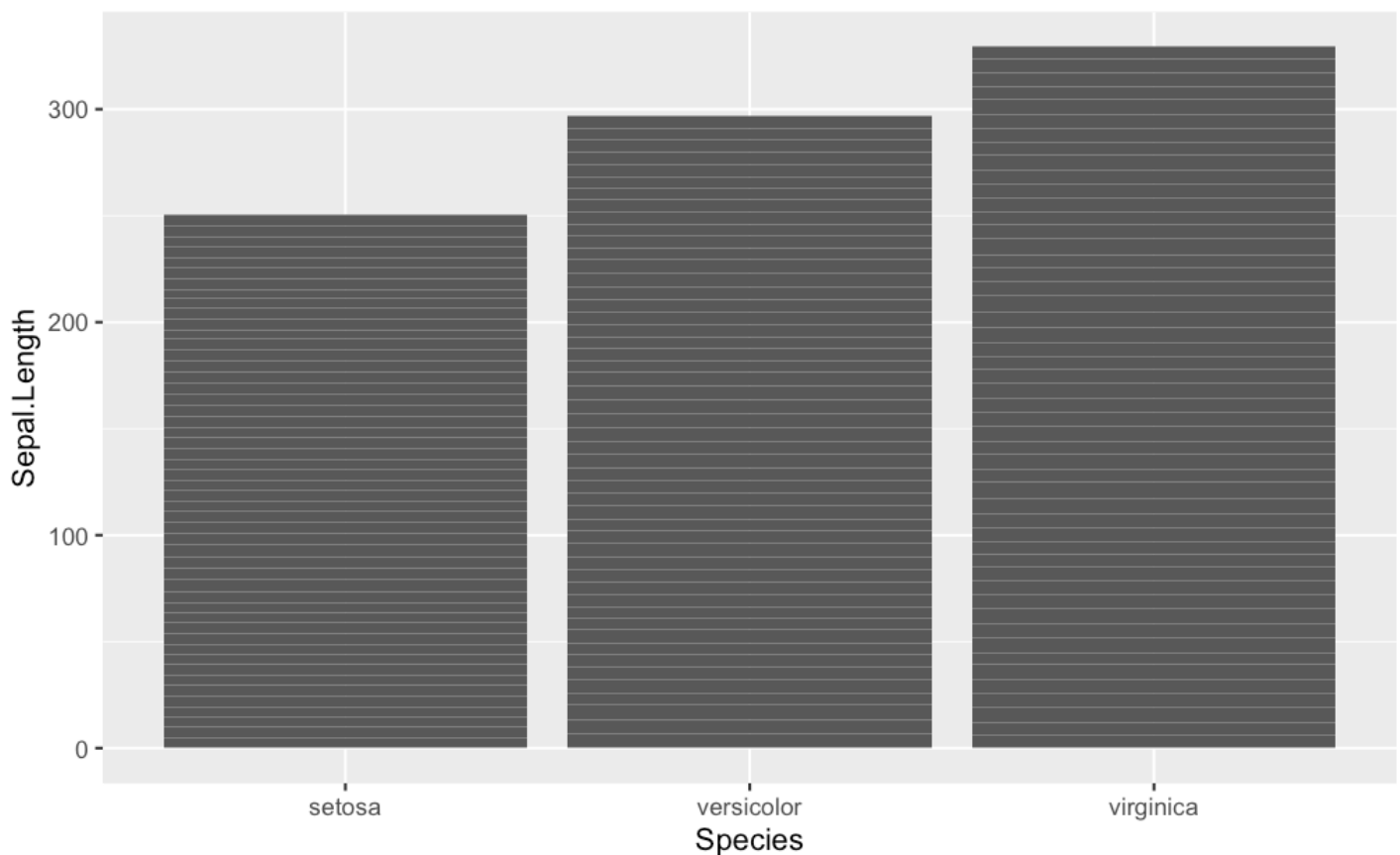
Hide

```
ggplot( iris, aes(x = Species, y = Petal.Length)) +  
  geom_col()
```



Hide

```
ggplot( iris, aes(x = Species, y = Sepal.Length)) +  
  geom_col()
```

[Hide](#)

```
ggplot(iris, aes(x = Sepal.Length, colour =Species)) +  
  geom_his() +  
  labs(  
    x = "Sepal lenght",  
    y = "Frequecy"  
  )
```

Error in geom_his() : could not find function "geom_his"

BOX PLOTS

The summary function is a quick and easy way to assess the statistical properties of each attribute. These values are displayed graphically in a box plot.

[Hide](#)

```
summary(iris[,1: 2])
```

Sepal.Length	Sepal.Width
Min. :4.300	Min. :2.000
1st Qu.:5.100	1st Qu.:2.800
Median :5.800	Median :3.000
Mean :5.843	Mean :3.057
3rd Qu.:6.400	3rd Qu.:3.300
Max. :7.900	Max. :4.400

[Hide](#)

```
summary(iris[,3: 4])
```

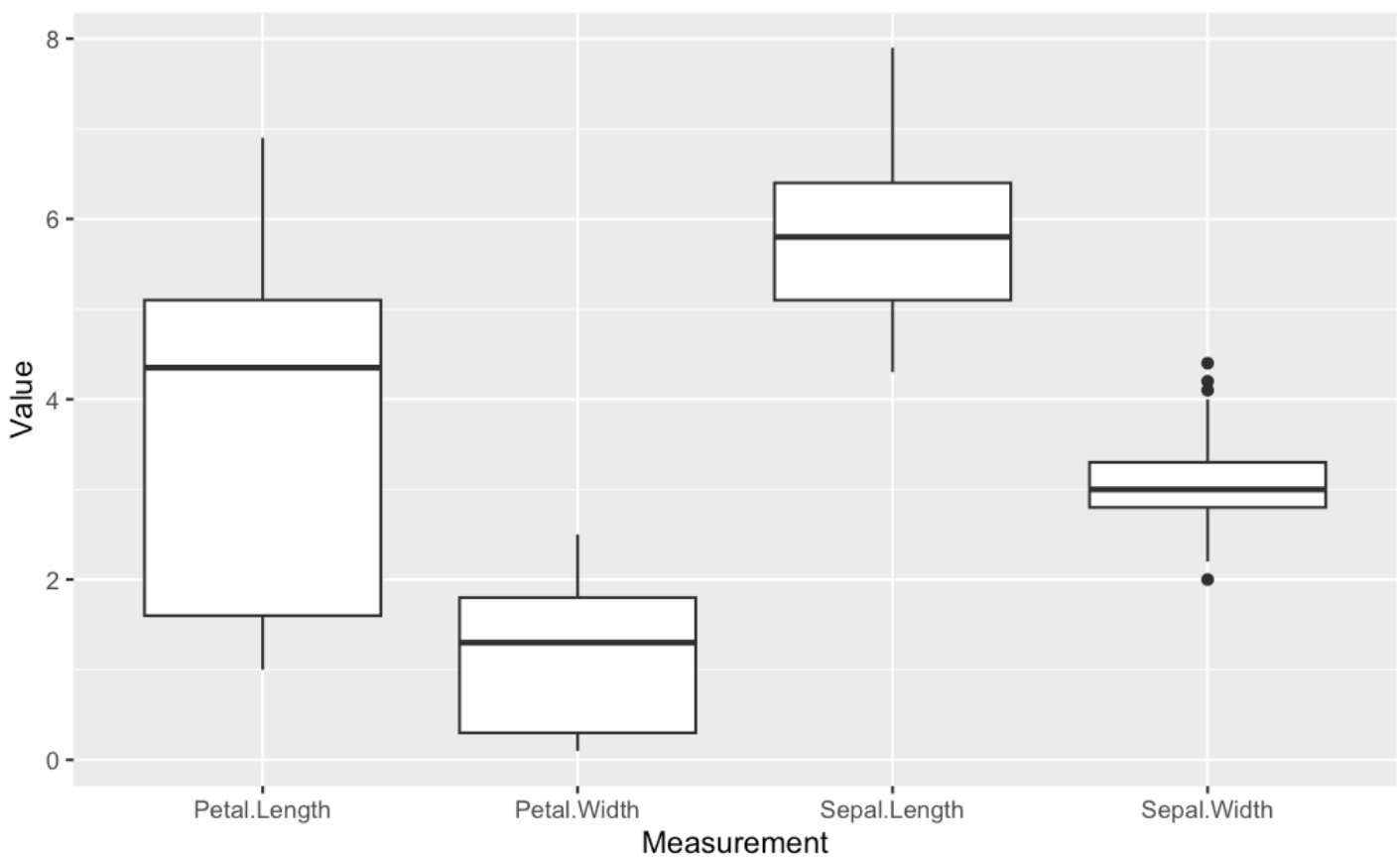
Petal.Length	Petal.Width
Min. :1.000	Min. :0.100
1st Qu.:1.600	1st Qu.:0.300
Median :4.350	Median :1.300
Mean :3.758	Mean :1.199
3rd Qu.:5.100	3rd Qu.:1.800
Max. :6.900	Max. :2.500

Box plots are used to compactly show many pieces of information about a variable's distribution. They are great for visualizing the spread of the data. Box plots show 5 statistically important numbers: the minimum, the 25th percentile, the median, the 75th percentile, and the maximum.

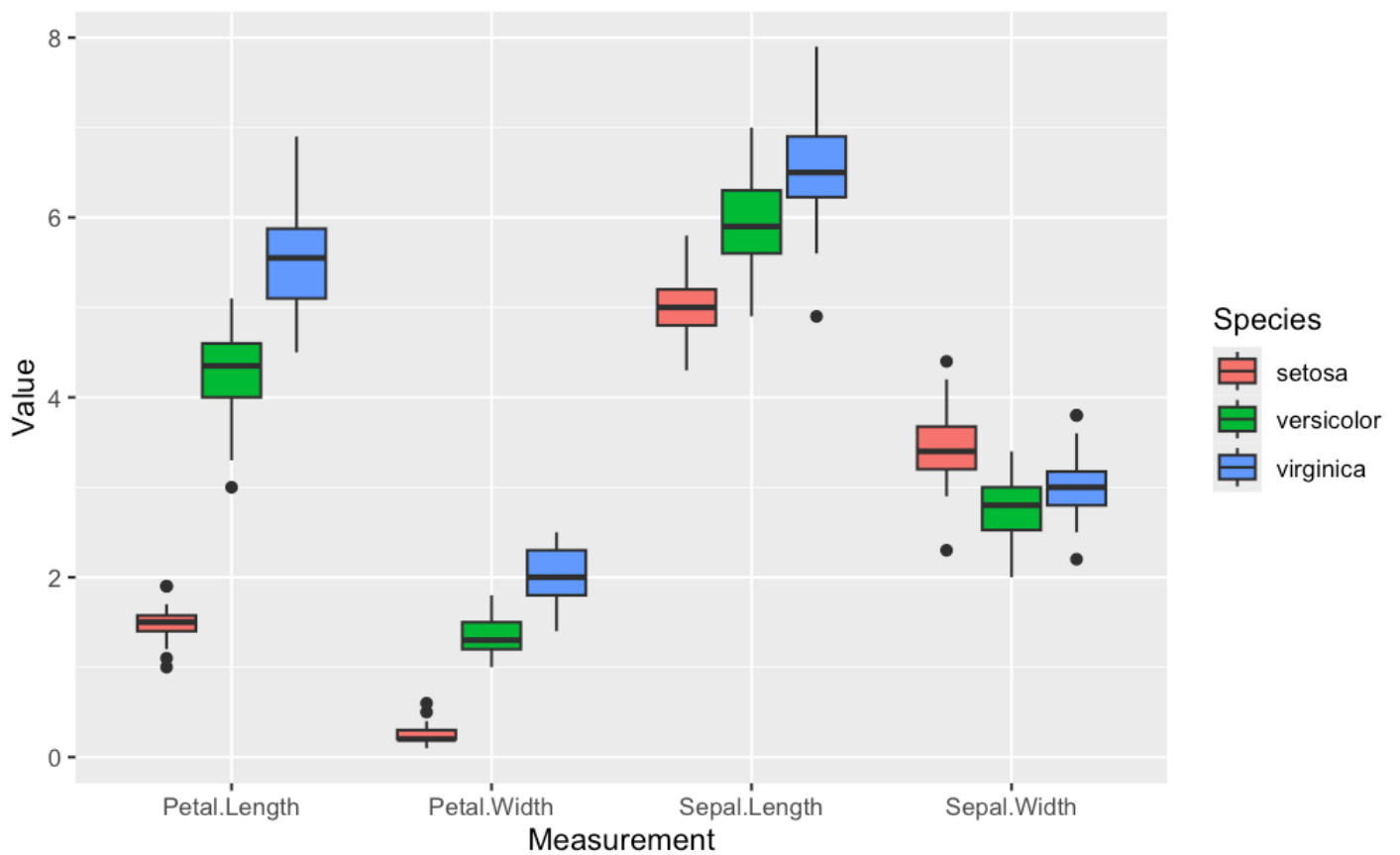
[Hide](#)

```
iris_long <- iris |> tidyr::pivot_longer(cols = c(Sepal.Length, Sepal.Width, Petal.Length, Petal.Width), names_to= "Measurement", values_to= "Value")

ggplot(iris_long, aes(x = Measurement, y = Value)) +
  geom_boxplot()
```

[Hide](#)

```
ggplot(iris_long, aes(x = Measurement, y = Value, fill = Species)) +
  geom_boxplot()
```

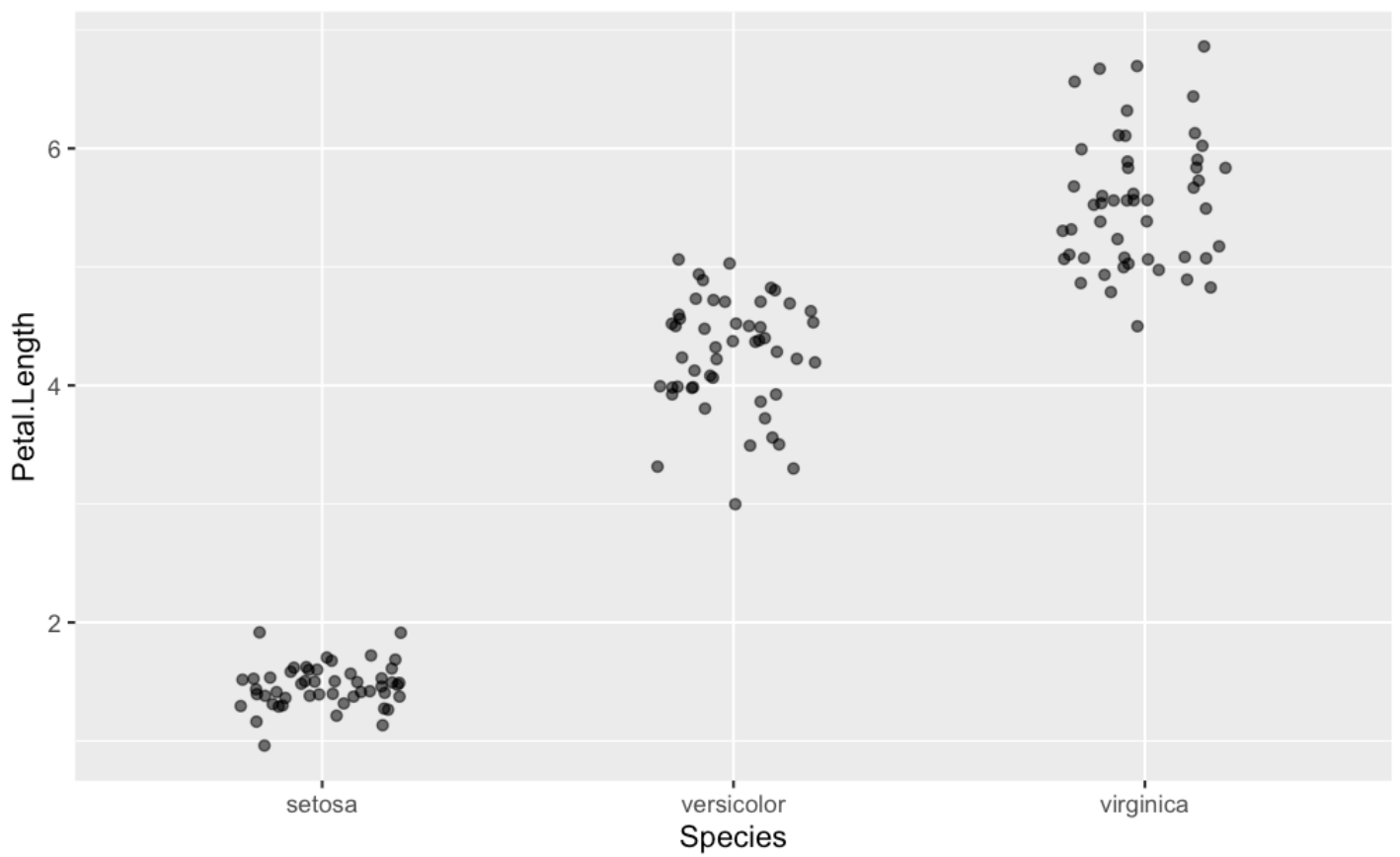
SCATTERPLOTS

Scatterplots are helpful for visualizing data and simple data inspection. Let's try the following code

Let's generate corresponding scatterplots for Petal.Width, Sepal.Length, and Sepal.Width.

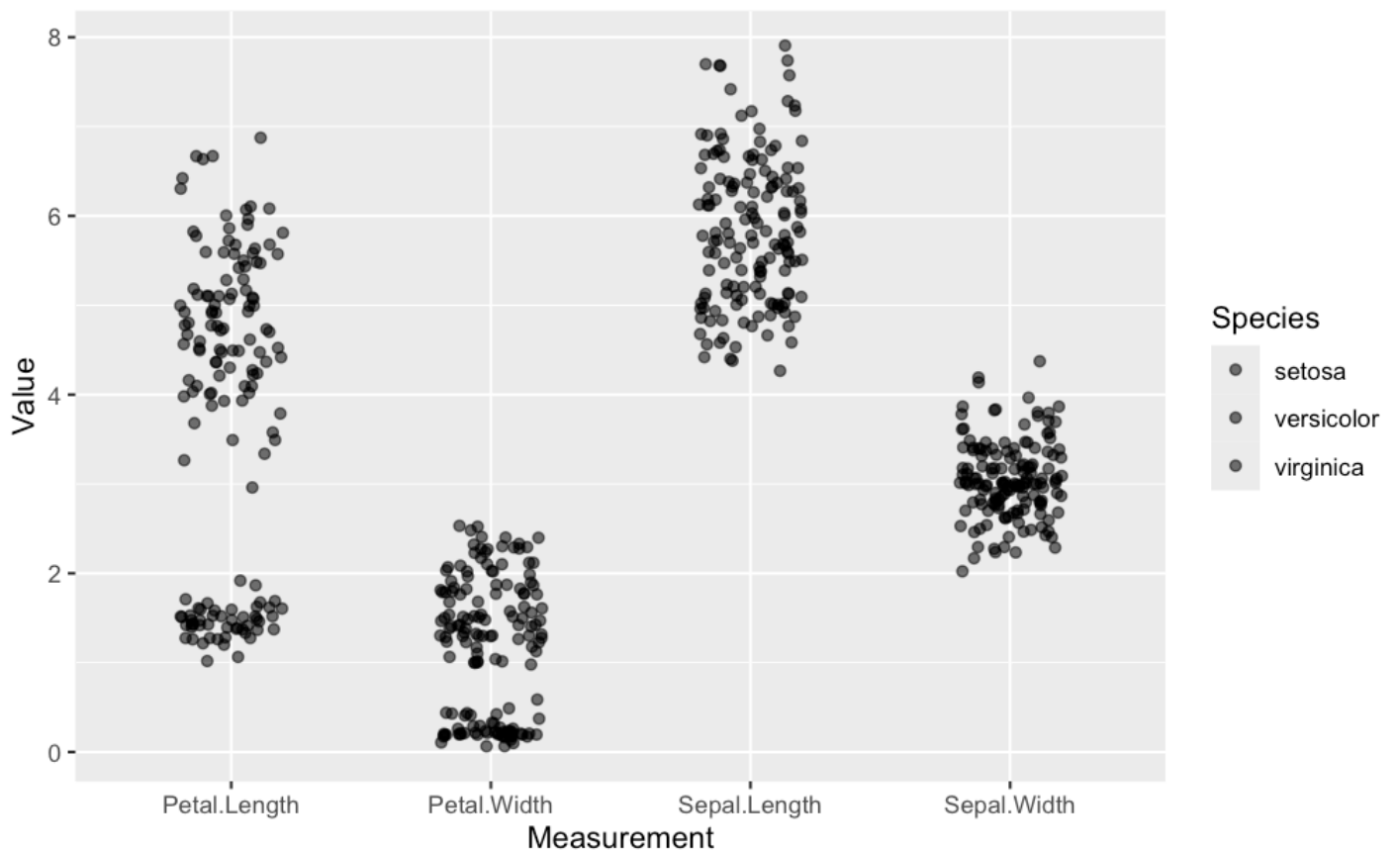
Hide

```
ggplot(iris, aes(x = Species, y = Petal.Length)) +  
  geom_point(alpha = 0.6, position = position_jitter(width = 0.2))
```



Hide

```
ggplot(iris_long, aes(x = Measurement, y = Value, fill = Species)) +  
  geom_point(alpha = 0.6, position = position_jitter(width = 0.2))
```



It continues in this fashion for Petal.Length and Petal.Width. Bring your attention to the bottom-left corner, the last plot in column 1. This plot has Sepal.Length on the x -axis and Petal.Width on the y -axis.

Hide

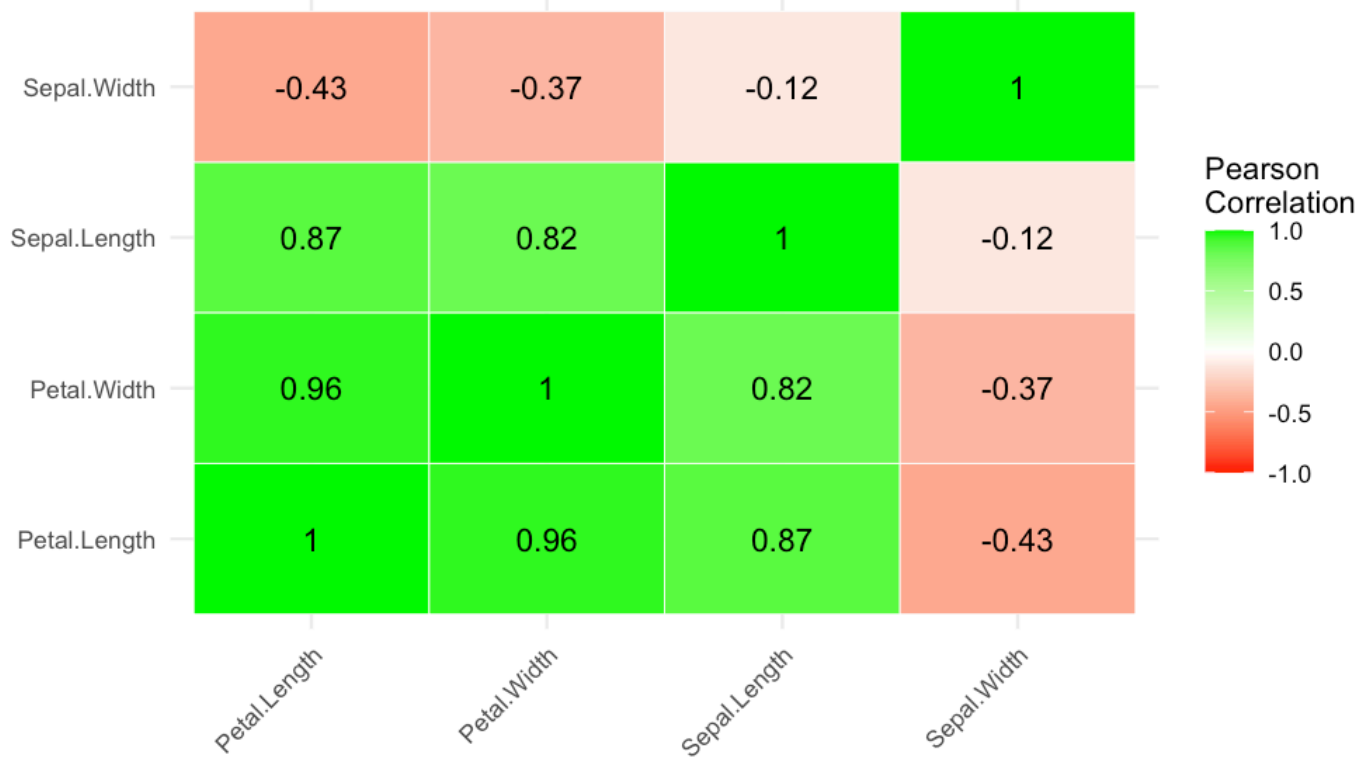
```
# 1. Calculate the correlation matrix
cor_matrix <- cor(iris[, 1:4])

# 2. Reshape the data for ggplot2
# First, convert the matrix to a data frame and add row names as a column
cor_df <- as.data.frame(cor_matrix)
cor_df <- tibble::rownames_to_column(cor_df, var = "Var1")

# Then, use pivot_longer() to reshape it
cor_long <- tidyr::pivot_longer(
  cor_df,
  cols = -Var1,
  names_to = "Var2",
  values_to = "Correlation"
)

# 3. Create the heatmap plot
ggplot(cor_long, aes(x = Var1, y = Var2, fill = Correlation)) +
  geom_tile(color = "white") +
  scale_fill_gradient2(
    low = "red", high = "green", mid = "white",
    midpoint = 0, limit = c(-1, 1),
    name = "Pearson\nCorrelation"
  ) +
  geom_text(aes(label = round(Correlation, 2)), color = "black", size = 4) +
  labs(
    title = "Correlation Matrix of Iris Measurements",
    x = "",
    y = ""
  ) +
  theme_minimal() +
  theme(axis.text.x = element_text(angle = 45, vjust = 1, hjust = 1))
```

Correlation Matrix of Iris Measurements



Hide

```
names(iris[, 1:4])
```

```
[1] "Sepal.Length" "Sepal.Width"  "Petal.Length" "Petal.Width"
```

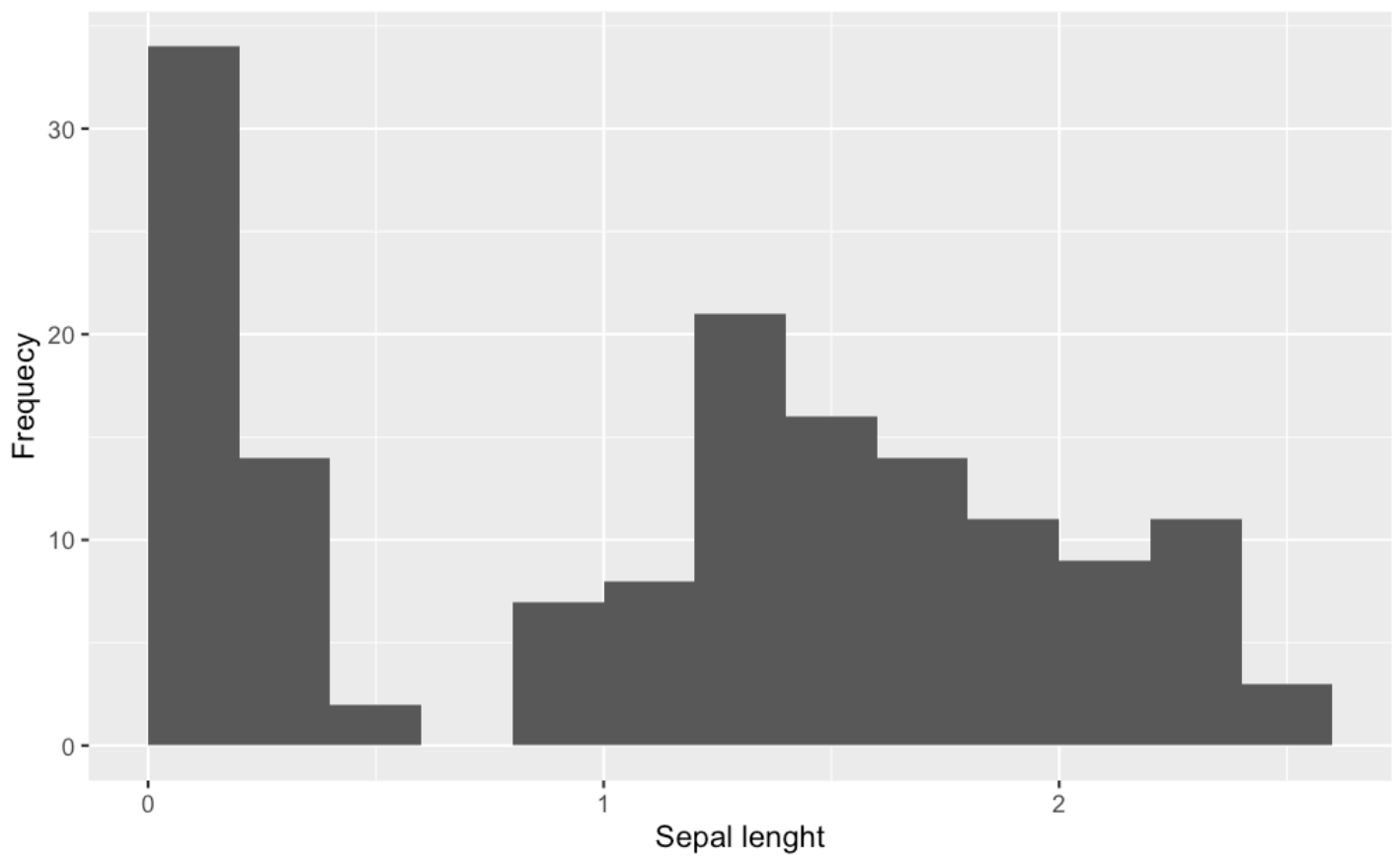
HISTOGRAMS

A histogram is a plot that breaks the data into bins (or breaks) and shows the frequency distribution of those bins.

Let's create some histograms of our iris data. The number of bins in the histogram is variable.

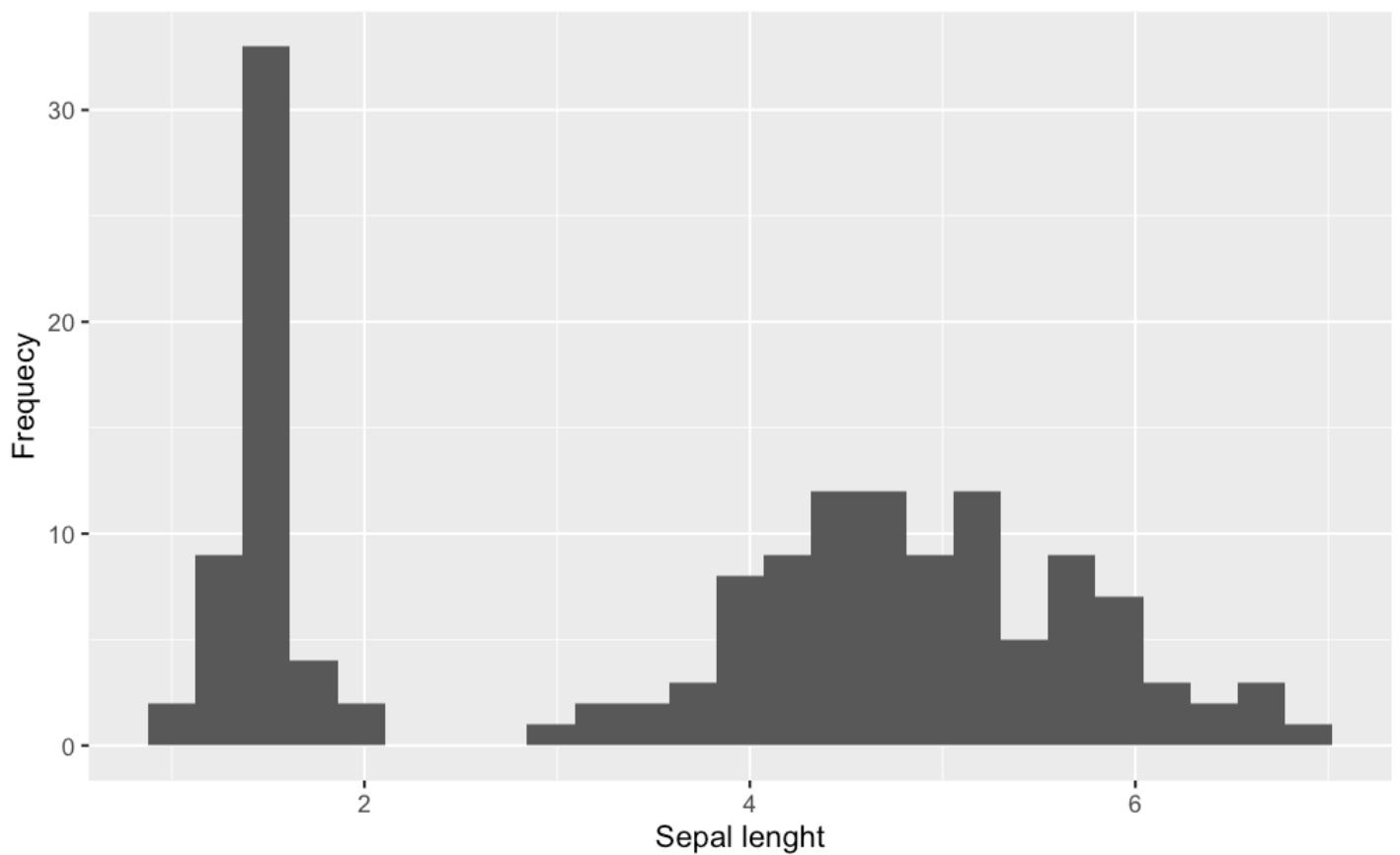
Hide

```
ggplot(iris, aes(x = Petal.Width)) +
  geom_histogram(bins = 13) +
  labs(
    x = "Sepal length",
    y = "Frequecy"
  )
```



Hide

```
ggplot(iris, aes(x = Petal.Length)) +  
  geom_histogram(bins = 25) +  
  labs(  
    x = "Sepal lenght",  
    y = "Frequecy"  
  )
```

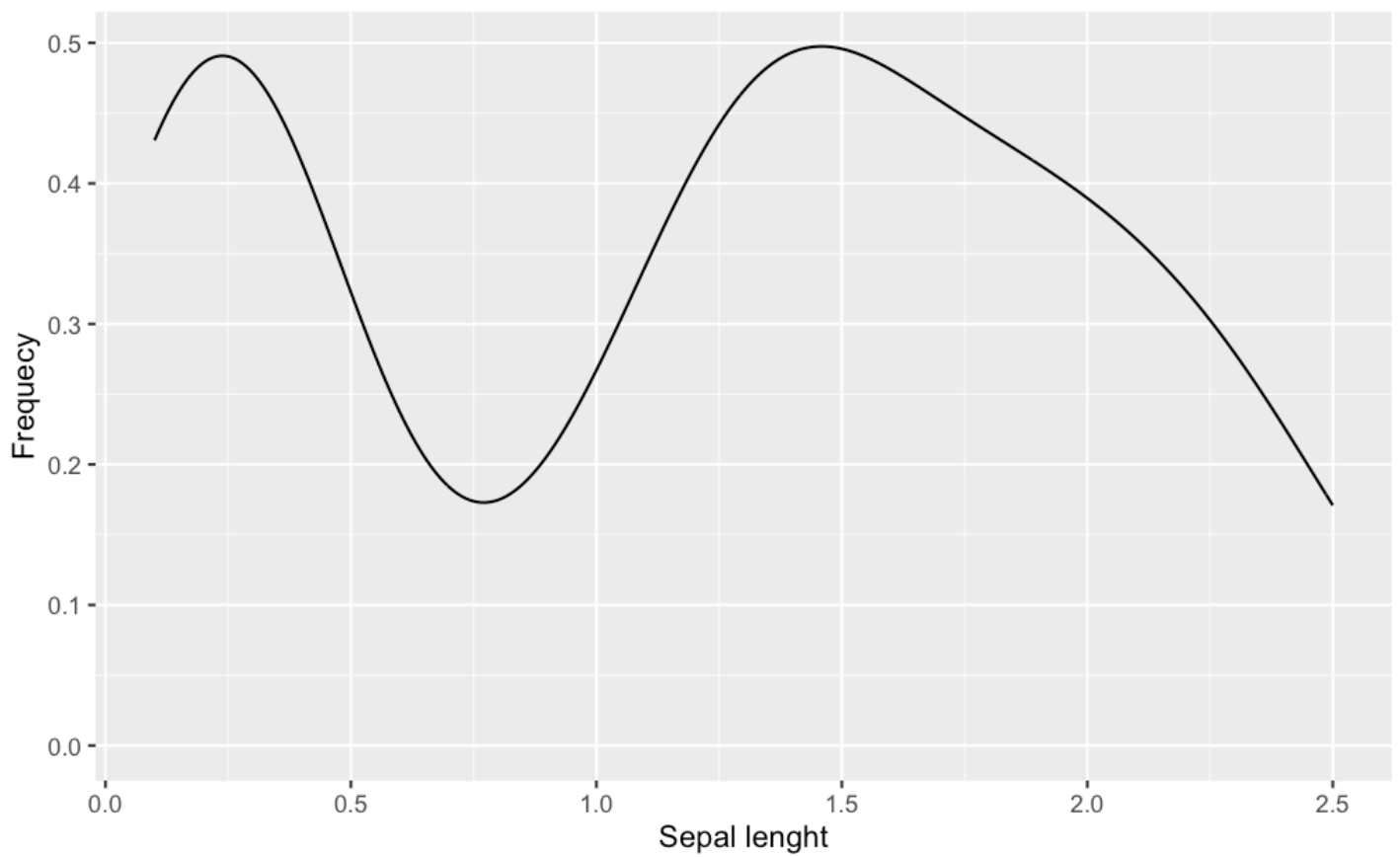


DENSITY PLOTS

Density plots can be viewed as smoothed versions of a histogram. We can estimate the density using R's density function.

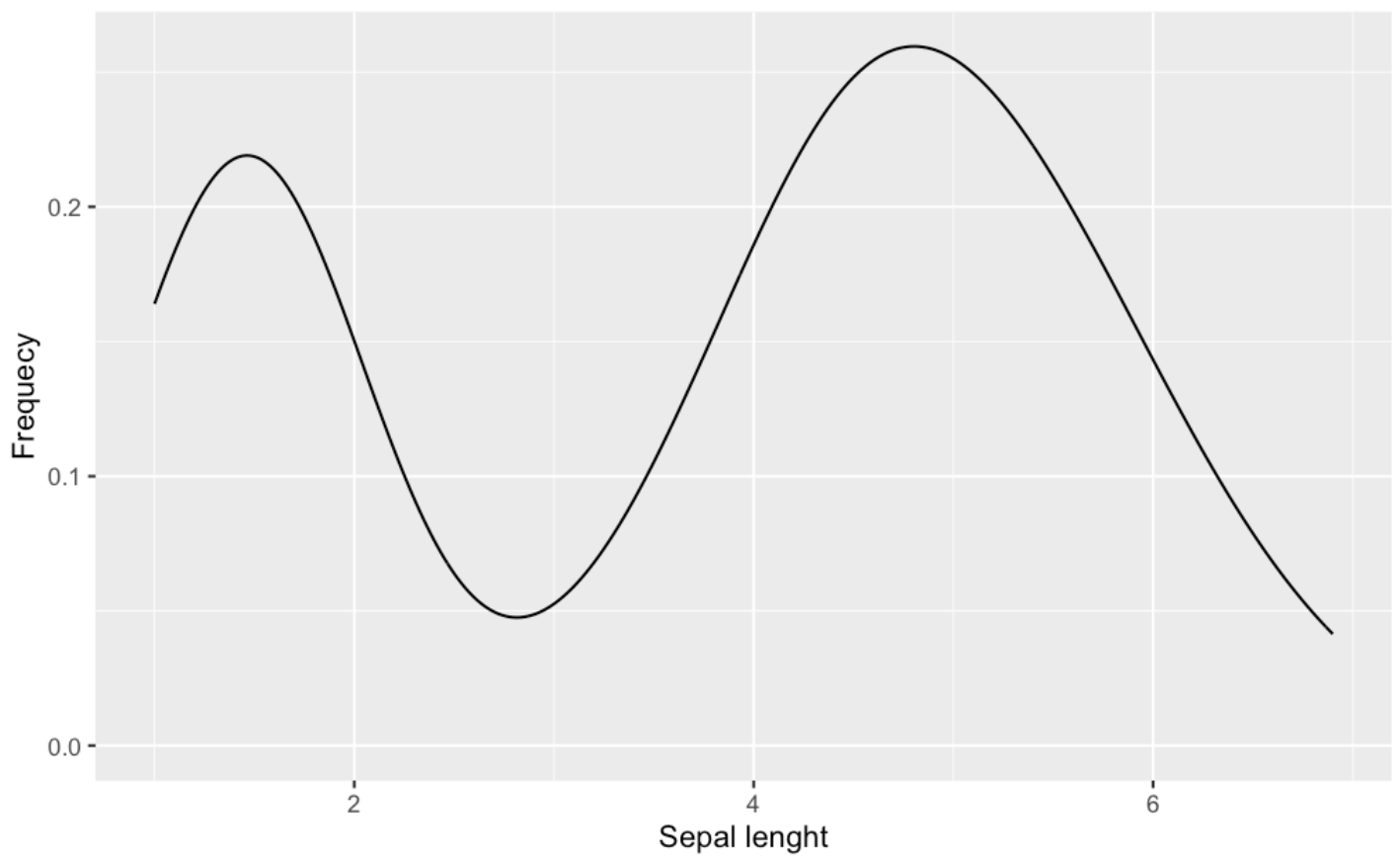
[Hide](#)

```
ggplot(iris, aes(x = Petal.Width)) +  
  geom_density() +  
  labs(  
    x = "Sepal lenght",  
    y = "Frequency"  
  )
```



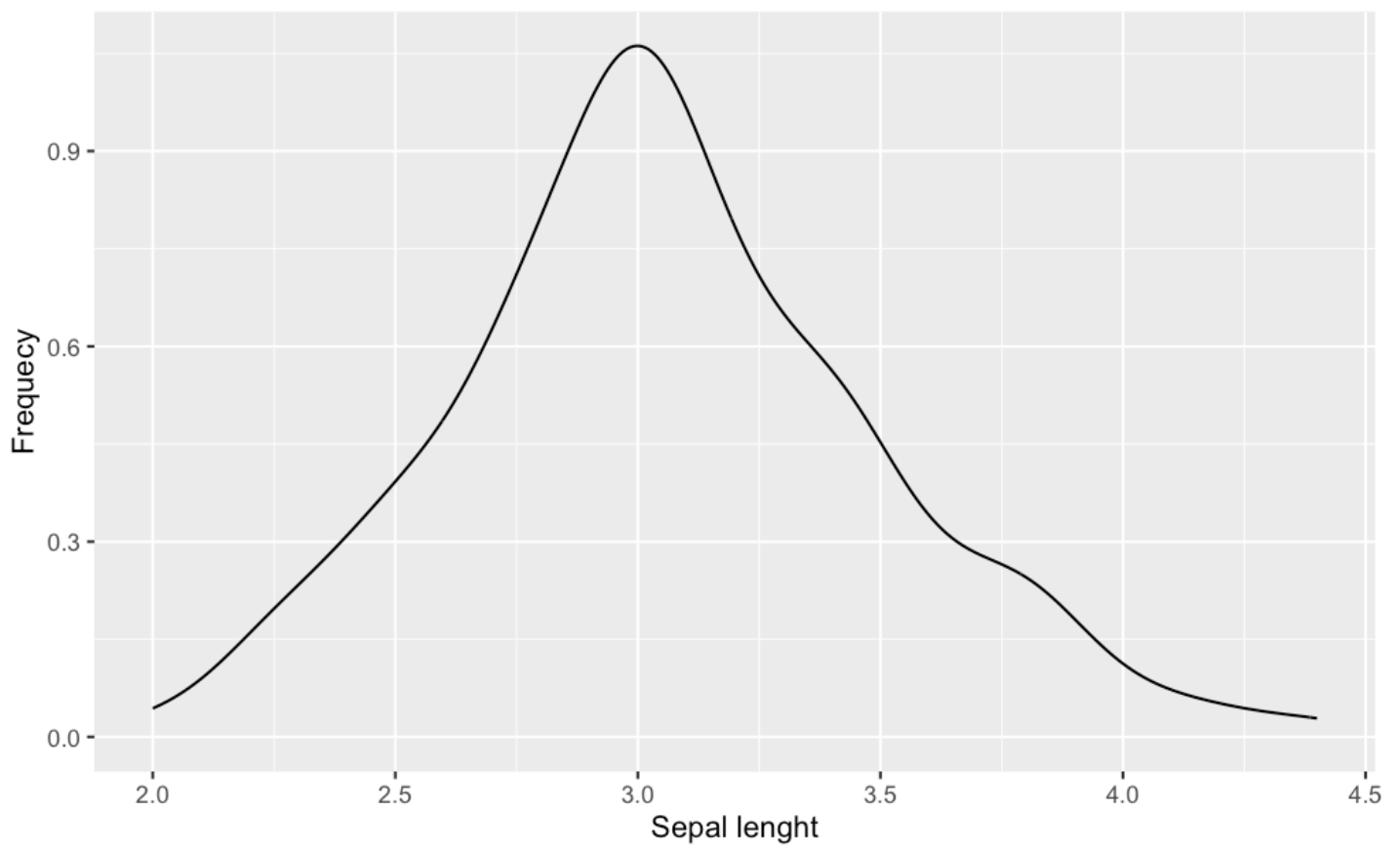
Hide

```
ggplot(iris, aes(x = Petal.Length)) +  
  geom_density() +  
  labs(  
    x = "Sepal lenght",  
    y = "Frequency"  
  )
```



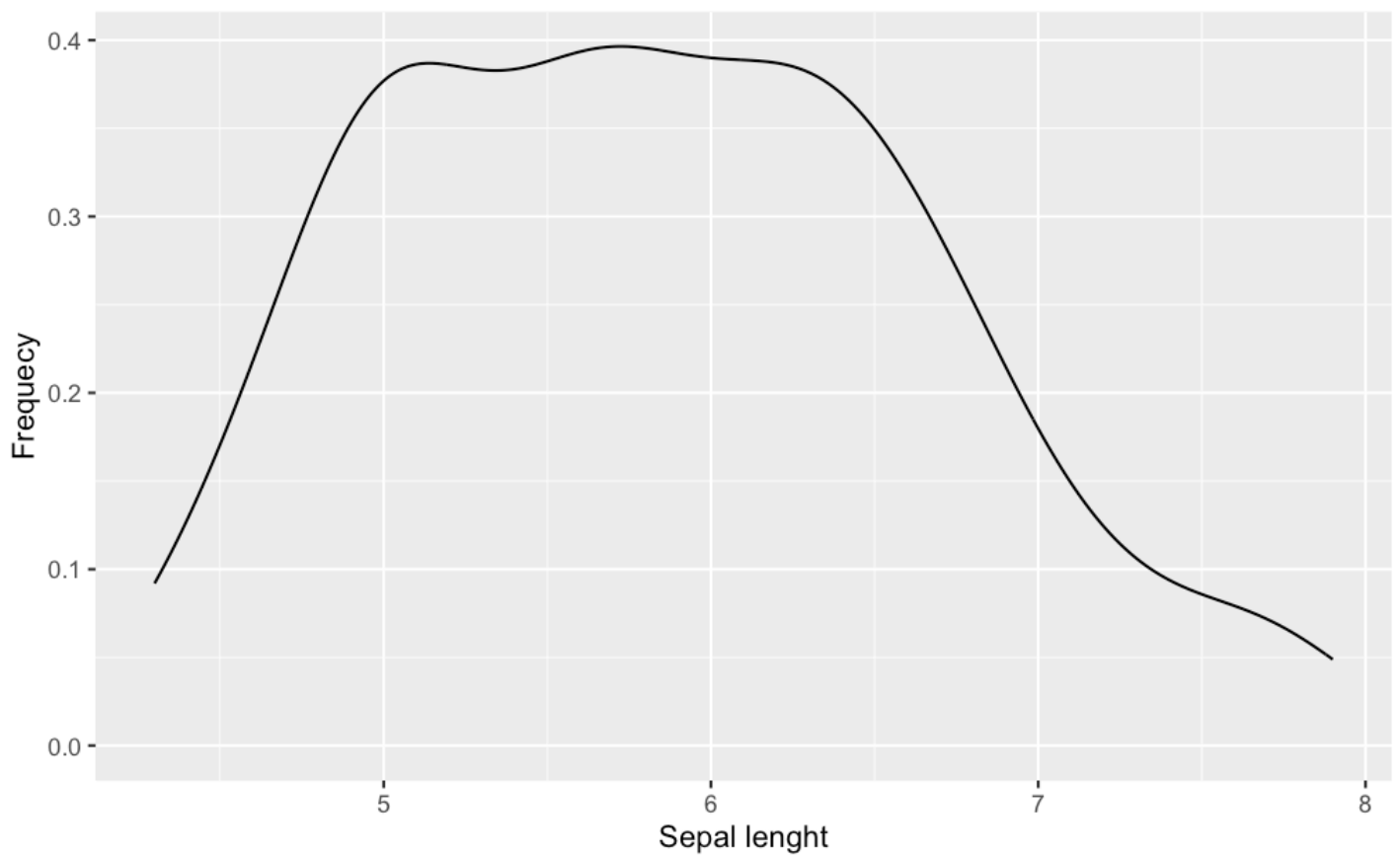
Hide

```
ggplot(iris, aes(x = Sepal.Width)) +  
  geom_density() +  
  labs(  
    x = "Sepal lenght",  
    y = "Frequecy"  
  )
```

Hide

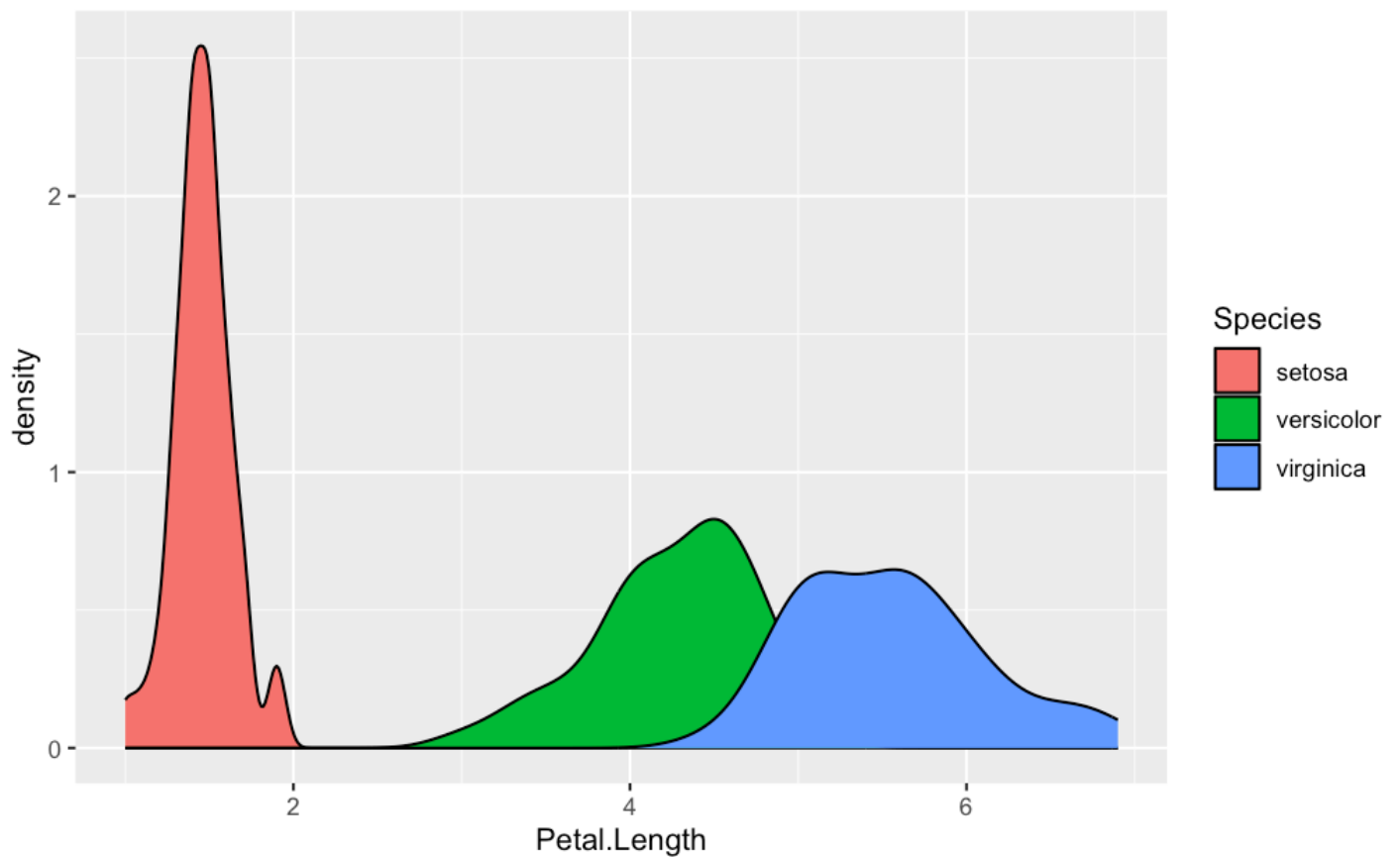
```
ggplot(iris, aes(x = Sepal.Length)) +  
  geom_density() +  
  labs(  
    x = "Sepal lenght",  
    y = "Frequecy"  
  )
```



Let's also look at the density function of Petal.Length for each of the 3 classes of irises.

Hide

```
ggplot(iris, aes(x = Petal.Length, fill = Species)) +  
  geom_density()
```



CONTOUR PLOTS

Density estimation is available for higher-dimensional data using contour plots. A contour plot is a graph that explores the potential relationship among 3 variables.

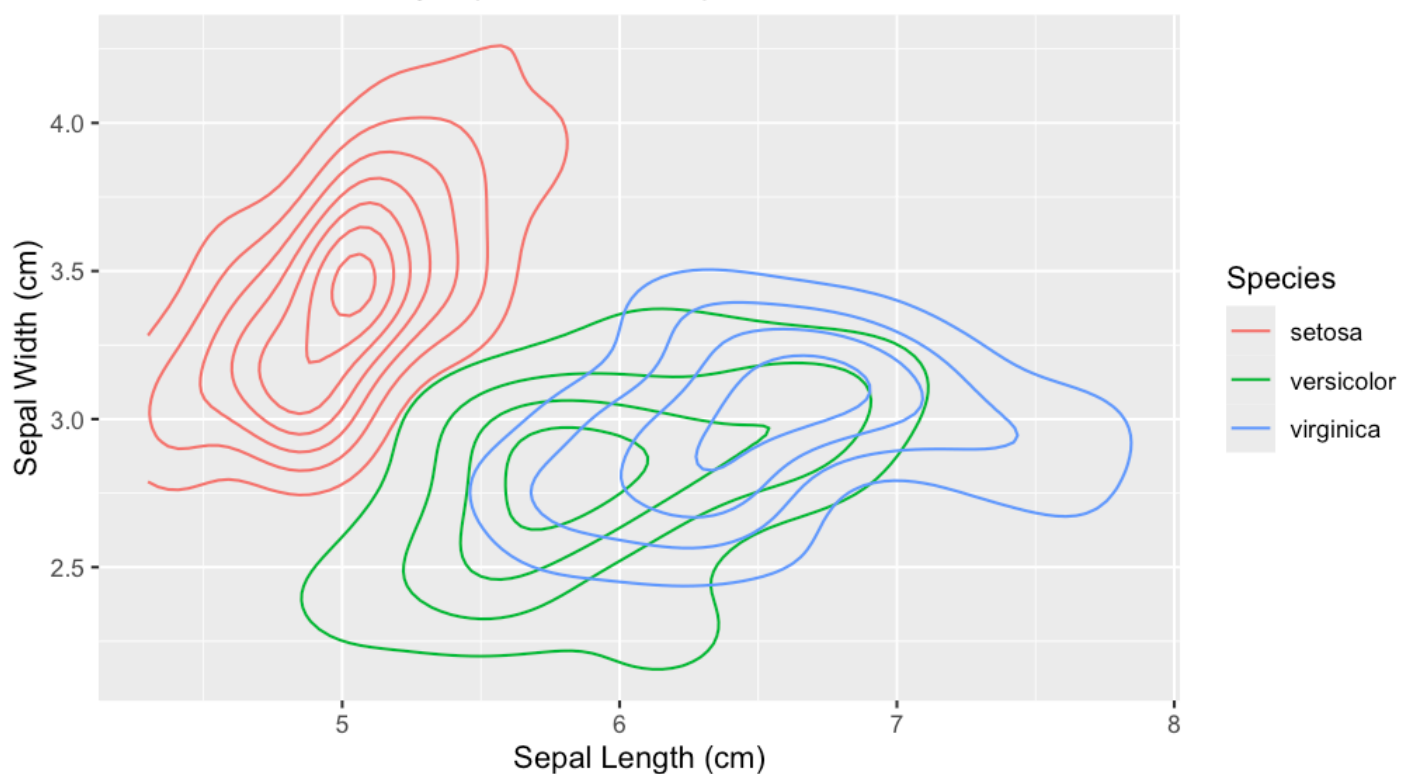
Contour plots display the 3-dimensional relationship in 2 dimensions, with x and y variables plotted on the x and y scales and the z variable represented by contours.

[Hide](#)

```
# Create the contour plot using geom_density_2d()
ggplot(iris, aes(x = Sepal.Length, y = Sepal.Width)) +
  geom_density_2d(aes(color = Species)) +
  labs(
    title = "Contour Plot of Iris Data Density",
    subtitle = "Traces show the density of points for each species",
    x = "Sepal Length (cm)",
    y = "Sepal Width (cm)"
  )
```

Contour Plot of Iris Data Density

Traces show the density of points for each species



The plot may also be viewed as a heat map, with brighter colors denoting more values in those regions.

[Hide](#)

```
# Create separate grid heatmaps for each species
ggplot(iris, aes(x = Sepal.Length, y = Sepal.Width)) +
  geom_density_2d_filled(aes(fill = after_stat(level))) +
  labs(
    title = "Grid Heatmap by Species",
    x = "Sepal Length (cm)",
    y = "Sepal Width (cm)"
  )
)
```

